

Loss Functions

Visium loss (MSE on soft cell-type proportions from RCTD):

$$\mathcal{L}_{\text{vis}} = \frac{1}{N_v} \sum_{i \in \mathcal{V}} \|\hat{p}_i - y_i\|^2 \quad (1)$$

Xenium loss (cross-entropy on hard single-cell labels):

$$\mathcal{L}_{\text{xen}} = -\frac{1}{N_x} \sum_{i \in \mathcal{X}} \log \hat{p}_{i, c_i} \quad (2)$$

CDAN domain adversarial loss with entropy conditioning:

$$\mathcal{L}_{\text{dom}} = -\frac{1}{N} \sum_{i=1}^N w_i \sum_{d=1}^D \mathbf{1}[d_i = d] \log \hat{d}_i^{(d)}, \quad w_i = 1 + \exp(-H_i) \quad (3)$$

where the domain classifier receives CDAN features:

$$f_i = \frac{\hat{p}_i \otimes z_i}{\sqrt{D}} \quad (4)$$

and a gradient reversal layer (GRL) negates gradients during backpropagation:

$$\text{GRL}(x) = x \text{ (forward)}, \quad \nabla \rightarrow -\lambda \nabla \text{ (backward)} \quad (5)$$

Reconstruction loss (masked MSE):

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{M}|} \sum_{j \in \mathcal{M}} \|\hat{x}_j - x_j\|^2, \quad \mathcal{M} \sim \text{Bernoulli}(r) \quad (6)$$

Total loss:

$$\mathcal{L} = \mathcal{L}_{\text{vis}} + \alpha \mathcal{L}_{\text{xen}} + \lambda(t) \mathcal{L}_{\text{dom}} + \rho \mathcal{L}_{\text{rec}} \quad (7)$$

Adversarial schedule:

$$\lambda(t) = \lambda_{\text{max}} \left(\frac{2}{1 + e^{-10p}} - 1 \right), \quad p = \frac{\text{epoch}}{\text{max_epochs}} \quad (8)$$

Multi-Phase Training

Phase	Aim	Data	Objective	Encoder Freezing
Phase 0	Reconstruction pretraining	All samples	$\mathcal{L}_{\text{rec}}$ only	None (full encoder trains)
Phase 1	Spot-data weak supervision	Spot-data only	$\mathcal{L}_{\text{vis}} + \mathcal{L}_{\text{dom}} + \mathcal{L}_{\text{rec}}$	First N layers frozen
Phase 2	Cell-data fine-tuning	Cell-data only	$\mathcal{L}_{\text{xen}} + \mathcal{L}_{\text{dom}} + \mathcal{L}_{\text{rec}}$	First $N-1$ layers frozen

Phase 0 — Reconstruction pretraining. Learns a general-purpose latent space from all H&E embeddings without any labels. The masked autoencoder objective forces the encoder to capture tissue morphology before seeing any supervision signal.

Phase 1 — Visium weak supervision. Trains on Visium spots with soft cell-type proportions from RCTD deconvolution ($\sim 55\mu\text{m}$ resolution). Partial encoder freezing preserves pretrained representations in early layers while adapting later layers.

Phase 2 — Xenium fine-tuning. Fine-tunes on Xenium single-cell annotations with hard labels at cellular resolution. Fewer layers are frozen ($N-1$ vs N) to allow slightly more adaptation, but the largely frozen encoder prevents catastrophic forgetting of Visium-learned features. CDAN domain adversarial loss aligns Visium and Xenium feature spaces across all phases.